

# Flight Operations Runbook — GCC-2024 Rev 1.0

## Gate Change Cascade Demo — Anomaly Response Procedures

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### §1 — Gate Wait Anomalies

A gate wait anomaly occurs when the average time passengers or aircraft spend waiting at a gate exceeds the ARIMA upper confidence bound by more than 95%.

#### Thresholds

| Condition                          | Severity |
|------------------------------------|----------|
| gate_wait > 30 min above baseline  | CRITICAL |
| gate_wait 15–30 min above baseline | HIGH     |
| gate_wait 8–15 min above baseline  | MEDIUM   |
| gate_wait < 8 min above baseline   | LOW      |

#### §1 Option A — Immediate Gate Reassignment (CRITICAL / HIGH)

**Trigger:** ARIMA anomaly score  $\geq 0.7$

**Action:** 1. Contact Hub Operations Center immediately (ORD: +1-312-555-0100 / DFW: +1-972-555-0200) 2. Request alternate gate assignment from Gate Control 3. Notify ground crew lead via radio channel ops-3 4. Update FIDS display; push passenger notification via app 5. If delay > 30 min, trigger passenger rebooking eligibility 6. Open ServiceNow incident (urgency 1) and assign to Gate Operations team

**Expected resolution:** 20–40 minutes

**Escalate to Option A+:** If no alternate gate available within 15 min, escalate to Airport Ops Director

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#### §1 Option B — Standby Monitoring with Ground Crew Dispatch (MEDIUM)

**Trigger:** ARIMA anomaly score 0.5–0.7

**Action:** 1. Dispatch ground crew supervisor to gate for visual inspection 2. Check jet bridge status in AIMS system 3. Monitor FIDS for delay cascade on connecting flights 4. If gate\_wait exceeds HIGH threshold within 15 min, escalate to Option A

**Expected resolution:** 15–25 minutes

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### §1 Option C — Monitor and Log (LOW)

**Trigger:** ARIMA anomaly score < 0.5

**Action:** 1. Log event in flight ops journal 2. No immediate crew dispatch required 3. Re-evaluate if anomaly persists for 3+ consecutive windows

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### §2 — Departure Delay Anomalies

A departure delay anomaly occurs when average departure delay deviates beyond the ARIMA confidence band, indicating a systemic delay pattern rather than normal operational scatter.

#### Delay Buckets

| Departure Delay | DOT Classification | Reporting Required           |
|-----------------|--------------------|------------------------------|
| < 15 min        | On Time            | No                           |
| 15–30 min       | Minor Delay        | Log only                     |
| 30–60 min       | Moderate Delay     | Notify passenger services    |
| > 60 min        | Significant Delay  | Full escalation + DOT report |

### §2 Option A — Full Escalation (CRITICAL / HIGH)

**Trigger:** departure\_delay value > 30 min AND anomaly\_score >= 0.7

**Action:** 1. Escalate to Duty Manager and inform Station Manager 2. Initiate passenger communication (app push + gate announcement) 3. Activate passenger rebooking protocol if delay > 60 min 4. Check crew rest compliance — if delay pushes crew over duty time limit, initiate crew swap 5. Coordinate with ATC for revised slot time 6. Open ServiceNow P1 incident; assign to Dispatch Operations

**Expected resolution:** 30–90 minutes

**Regulatory note:** FAA tarmac delay rule (3h domestic / 4h international) must be monitored

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### §2 Option B — Proactive Notification (MEDIUM)

**Trigger:** departure\_delay 15–30 min AND anomaly\_score 0.5–0.7

**Action:** 1. Push delay notification to passengers via app 2. Update gate agent on revised estimated departure 3. Coordinate gate hold with jetway crew 4. Monitor connecting passengers — flag minimum connection time risks

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## §2 Option C — Monitor (LOW)

**Trigger:** anomaly\_score < 0.5

**Action:** 1. Log in daily ops report 2. Gate agent monitors and reports if delay increases

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## §3 — Turnaround Time Anomalies

A turnaround anomaly occurs when average aircraft turnaround time exceeds the ARIMA forecast, indicating ground operations are running longer than expected.

### Turnaround Root Cause Checklist

| Root Cause            | First Check                                     |
|-----------------------|-------------------------------------------------|
| Fueling delay         | Contact fuel vendor — check truck availability  |
| Catering delay        | Contact catering coordinator                    |
| Cleaning delay        | Check cleaning crew headcount vs. aircraft size |
| Baggage loading delay | Check belt loader availability at gate          |
| Late inbound aircraft | Downstream impact — check block-out time        |
| Maintenance hold      | Coordinate with Line Maintenance (see §4)       |

## §3 Option A — Ground Operations Escalation (CRITICAL / HIGH)

**Trigger:** turnaround\_time value > 60 min OR anomaly\_score >= 0.7

**Action:** 1. Dispatch Ramp Supervisor to gate immediately 2. Conduct root cause checklist above; address highest-impact item first 3. Request priority fueling and/or catering if vendor SLA is breached 4. If cleaning is root cause: authorize premium cleaning crew overtime 5. Update block-out estimate in AIMS; notify Crew Scheduling 6. Open ServiceNow incident (urgency 1); assign to Ground Operations

**Expected resolution:** 20–45 minutes

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## §3 Option B — Expedited Coordination (MEDIUM)

**Trigger:** turnaround\_time 15–20 min over baseline AND anomaly\_score 0.5–0.7

**Action:** 1. Contact gate coordinator; request status of each ground service 2. Prioritize fueling completion (critical path) 3. Authorize catering team to begin boarding before catering complete if items are non-essential

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### §3 Option C — Monitor (LOW)

**Trigger:** anomaly\_score < 0.5

**Action:** 1. Log in daily ops report 2. No immediate escalation needed

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### §4 — Fleet Reference

| Aircraft | Type     | Hub | Flights             | Ops Contact                         |
|----------|----------|-----|---------------------|-------------------------------------|
| N101AX   | B737-800 | ORD | AX101<br>(ORD->LAX) | ORD Station<br>Ops: +1-312-555-0110 |
| N202AX   | A320neo  | ORD | AX202<br>(ORD->JFK) | ORD Station<br>Ops: +1-312-555-0110 |
| N303AX   | B737-MAX | DFW | AX303<br>(DFW->MIA) | DFW Station<br>Ops: +1-972-555-0210 |
| N404AX   | A321XLR  | DFW | AX404<br>(DFW->SEA) | DFW Station<br>Ops: +1-972-555-0210 |

### Hub Operations Centers

| Hub | Phone           | Slack Channel   | ServiceNow Assignment Group |
|-----|-----------------|-----------------|-----------------------------|
| ORD | +1-312-555-0100 | #ops-ord-alerts | Gate Operations ORD         |
| DFW | +1-972-555-0200 | #ops-dfw-alerts | Gate Operations DFW         |

### Escalation Chain

Gate Agent

    +-- Gate Coordinator

        +-- Duty Manager / Station Manager

            +-- Hub Operations Director

                +-- VP Flight Operations (CRITICAL only)

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*For demo purposes only. Contact information is fictional.*